AIDAN GIEBELHAUSEN

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AEROSPACE SYSTEMS ENGINEER | Control Systems | CFD | Guidance & Navigation | MATLAB & Simulink

EDUCATION

PURDUE UNIVERSITY **GPA: 3.22/4.0**

BACHELOR OF SCIENCE IN AERONAUTICAL AND ASTRONAUTICAL ENGINEERING

WORK EXPERIENCE

F-35 SYSTEMS ENGINEER, LOCKHEED MARTIN **2026-PRESENT**

Accepted offer; awaiting Secret security clearence.

AEROSPACE ENGINEER, BAXTER AEROSPACE **2025-2026**

Designed the full aerodynamic configuration for the Dragonfly UAV, including wing planform, control surfaces, high-lift devices, and stabilators. Developed a parametric aerodynamic analysis pipeline integrating AVL, XFOIL, and Python to automate geometry and flight-condition sweeps, and built validated AVL models to assess stability derivatives, control authority, trim, and flight-envelope behavior. Designed a 3-DOF longitudinal autopilot architecture, implementing the Dragonfly-specific plant model and trim-state solver. Performed servo characterization and system-level testing for a 6-DOF thrust-vectoring propulsion system, confirming angular response, authority margins, and repeatability. Designed and fabricated the Dragonfly electrical power and wiring harness, integrating the flight computer, servo power board, battery system, servos, wingtip nodes, RTK GPS, radar, and LiDAR using rapid harness prototyping with multi-gauge wiring, soldered multi-way splices, ring terminals, and Harwin and Molex connectors. Developed embedded flight software using MCUXpresso, implementing RTK GPS and LiDAR interfaces on the flight computer and creating a centralized state estimator leveraged across firmware components to provide real-time vehicle state awareness. Supported construction of the mobile command center trailer, installing RTK GPS base-station equipment, camera systems, and communications infrastructure, and contributed to design trade studies providing data-driven recommendations on aerodynamic performance, stability, control, and system integration.

CONTROL & SYSTEMS GRADUATE RESEARCHER, PURDUE UNIVERSITY **2024-2025**

Designed and implemented a convex optimization program to model and solve a spacecraft soft capture problem with safety-critical constraints. The project focused on optimizing autonomous docking maneuvers under thrust, velocity, and field-of-view limitations. Developed and fine-tuned algorithms using MATLAB and convex optimization software to ensure precise control and safe docking procedures. Conducted extensive simulations to validate the optimization program and assess its performance under various scenarios. Authored a comprehensive 30-page research paper detailing the methodology, results, and implications of the study. Presented findings to a leading astrodynamics expert, Kenshiro Oguri, at Purdue University.

PROJECT EXPERIENCE

CESSNA AUTOPILOT SYSTEM **2024-2025**

Designed and simulated a multi-axis autopilot in MATLAB, MATLAB Simulink, and Python for a Cessna 172 model. The project involved creating a detailed aircraft model, developing control algorithms for pitch, roll, and yaw, and integrating a Kalman filter to reduce sensor noise. The PID control algorithms were fine-tuned through iterative testing and optimization. Performance validation was conducted through a series of simulated flight tests, including step response analysis and stability assessments under various flight conditions.

SYSTEMS ENGINEERING AND CONTROL SYSTEMS LAB **2024-2025**

Worked with a Quanser Cube stabilization platform and fan-driven stability system to design and test control algorithms. The project involved applying system identification techniques to develop accurate models of the system dynamics and tuning PID controllers to achieve desired performance. The control algorithms were validated through experimental testing, and the models were refined based on the test results. Gained insight into feedback control, system integration, and the translation of theoretical models into practical hardware performance.

CRYOVENTING PROJECT TEAM LEAD (NASA / ZERO G) **2024-2025**

Led a 4-person design team to build a liquid nitrogen-based test system for microgravity flight. The project involved coordinating mechanical design, modeling and preproduction of system, and overseeing the fabrication process. Demonstrated hands-on experience in systems integration, safety-critical hardware, and testing logistics in challenging conditions. Developed detailed thermal and P&ID sketches to be implemented in the cyrogenic system, as well as CAD models in NX and Solidworks and CFD simulations in ANSYS to ensure the system's performance and safety. Findings and results were presented to NASA in Spring of 2025.

HOMELAND INTERCEPTOR PROJECT MANAGER

Led a 10-person design team through conceptual sizing and full design of a high-speed missile interceptor for an AIAA national defense competition. The project emphasized supersonic aerodynamics, control surface design, and already produced products to increase cost efficiency in high impact systems. Directed CFD analysis, propulsion matching, and system trade studies to meet mission-level objectives under time-critical constraints. Coordinated the development of detailed CAD models to ensure accurate system representation and integration. Conducted extensive simulations and performance evaluations using ANSYS Fluent, Openvsp, and XFLR5 to optimize aerodynamic performance and thermal management. Presented the final design and findings to Lockheed Martin. **2024-2025**

HYPERSONIC CFD RESEARCH

Conducted a comprehensive CFD study on an F-16 converging-diverging nozzle using ANSYS Fluent. The project involved investigating five different operating conditions, with detailed post-processing performed in MATLAB. Gained Experience in modeling gas flow, thermal gradients, and boundary condition effects in mechanical systems. The study included generating and refining computational grids, performing simulations to evaluate flow characteristics, and assessing mesh convergence. Conducted flow visualization to validate the aerodynamic performance and boundary condition behavior. Presented findings to a leading hypersonic expert, Gregory Blaisdell, at Purdue University. **2024-2025**

NASA STUDENT LAUNCH

Created a CAD Model in Solidworks for flight on a NASA Student Launch rocket. The project involved 3D printing multiple pieces for interconnected research projects, including a test for a new natural connection adhesive and a new microscopic set of accelerometers. Developed and implemented the electronic and camera setup for the testing equipment on the rocket. This hands-on experience provided valuable insights into CAD modeling, 3D printing, and the integration of electronic systems for aerospace applications. **2024-2025**

SPACEPORT CUP

Worked on the design, development, and integration of electronic systems for collecting and transmitting flight parameter data during rocket launches. Produced printed circuit boards (PCBs) and created and validated code to be run onboard the rocket. Collaborated with multidisciplinary teams to ensure the reliable operation of hardware and software systems in high-stress flight environments. This project emphasized the importance of teamwork, system integration, and the ability to work under pressure. **2024-2025**

HIGH POWERED ROCKETS TEAM

Contributed to successful Level 1 certification through the design, construction, and launch of high-powered rockets. Independently designed, built, and launched a Class H rocket, gaining hands-on experience in propulsion, structural assembly, and avionics integration. Supported testing and launch operations to refine the design and ensure mission success. This experience highlighted the importance of attention to detail and safety protocols. **2023-2025**

AAE 412 CFD AIRFOIL ANALYSIS

Completed an advanced aerodynamics project involving the analysis of airflow over an airfoil using MATLAB, Fortran, and ANSYS Fluent. Responsibilities included generating and refining computational grids, performing simulations to evaluate lift and pressure distribution, and assessing mesh convergence. Conducted flow visualization to validate aerodynamic performance and boundary condition behavior.

AERODYNAMICS AND BOEING WIND TUNNEL LAB WORK **2024-2025**

Illustrated various aerodynamic phenomena including airfoil pressure distribution, finite wing lift, drag, and moment, as well as supersonic flow around bodies and shock waves. Conducted experiments in a full-scale Boeing wind tunnel at the Purdue Airport.

SKILLS

- **Optimization & Control**: Experience with convex programming for system optimization and control applications, Model Based Systems Engineering tools.
- **Programming & Scripting**: Skilled in MATLAB, Simulink, Python, Java, JavaScript, C++ and C
- **CFD Tools**: Proficient in ANSYS Fluent and Fortran for computational fluid dynamics and flow simulation.
- **Modeling & Design**: Experienced in Autodesk Inventor, Siemens NX, and SolidWorks, Hypersonic work.
- **Numerical Methods**: Knowledge of finite element and finite difference methods; applied in fluid analysis.
- **Data Analysis & Tools**: Advanced proficiency in Microsoft Excel, LabVIEW, and TestStand Software, Open Mission Systems, developing embedded real-time software.
- **Teamwork & Leadership**: Led small engineering subteams in student research and projects, coordinating design reviews and task delegation. Collaborated across multidisciplinary groups on rocket design, flight data systems, and control projects.
- **Software and Devlopment**: Github, Azure devops, and MCUExpresso SDK.